Goirik Chakrabarty

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Summary

I am a Data Science Ph.D. candidate specializing in scaling transformer models to map visual inputs to high-dimensional neuronal activity (signals). My research focuses on improving stability and generalization when analyzing this noisy time-series data. While primarily using PyTorch, JAX, and Numba. Recently, I have started expanding into 3DGS simulation using Blender, Unity, and PyTorch3D to render experiment environments (mazes) and model corresponding neuronal responses from a mouse in it.

Experience

PhD Candidate, Data Science — Oct 2024 – Oct 2027
Advisor: Prof. Fabian Sinz — *Göttingen, Germany*

- **Designing Scalable Multimodal Transformers:** Architecting masking strategies for large-scale foundation models capable of mapping complex visual inputs to high-dimensional representation spaces.
- **High-Performance Video Encoding:** Leveraged Flash Attention 2 and Distributed Data Parallel (DDP) to optimize training pipelines for video inputs, significantly increasing training throughput on GPU clusters.
- **3D Neural Simulation:** Utilizing 3DGS (3D Gaussian Splatting), Blender, and Unity to architect differentiable pipelines for rendering complex 3D environments.
- **Foundation Model Evaluation:** Quantifying generalization performance and scaling laws as a function of stimulus entropy within diverse, large-scale data mixtures.

Pre-Doctoral Researcher — Jun 2023 – Sep 2024
TCS Innovation Hub — *New Delhi, India*

- **Multimodal Foundation Models**: Conducted deep-dive research into **CLIP** and **diffusion-based architectures** for advanced image manipulation and zero-shot editing tasks.
- **Generative Safety & Editing**: Developed **ReMOVE** (CVPRW 2024), a reference-free metric for evaluating concept erasure in generative modeling.
- **High-Efficiency Diffusion**: Designed **LoMOE** (ACMMM 2024), a localized multi-object editing framework achieving **50% faster inference** than state-of-the-art models through single-pass editing.
- **Product-Scale Prototyping**: Developed proof-of-concept generative pipelines for enterprise-scale e-commerce catalog generation.

Master's Thesis: Test-Time Domain Adaptation — Jul 2022 – May 2023
Indian Institute of Science (IISc) — *Bangalore, India*

- Implemented multiple adaptation methods (MixStyle, Prototype Classifier, Regularization techniques).
- Published a plug-and-play Simple signal for Domain Shift that converts any TTA method to CTTA, improving **ImageNet-C accuracy by 3.2%** with *smaller, faster models.*
- Published **SANTA**, which recognised the importance of *long-term stability* for continual test time adaptation and was better than SoTA by **9.6% on ImageNet-C**.

Machine Learning and Deep Learning Lead — Jun 2022 – Oct 2022
Curem Biotech (Startup) — *Pune, India*

- Built on device **Random Forest** and **XGBoost** based cell segmentation models to classifying parasitic and healthy RBCs.
- Designed **YOLO**-based pipeline for malaria species detection on these images after malaria detection.

Skills

Machine Learning / Deep Learning: PyTorch, HuggingFace, PyTorch Lightning, Distributed Data Parallel (DDP), TensorFlow, JAX, Scikit-learn, NumPy, Numba, MLOps
Programming Languages: Python (expert), MATLAB (intermediate), R (intermediate)
Data Science / Research: Multimodal Learning, Diffusion Models, GANs, Test-Time Adaptation, Domain Adaptation
Tools & Frameworks: Git, GitHub, Docker, Apptainer, SLURM, WandB, MS Excel, Linux/Command Line, MPI, OpenMP
Web / Typesetting: LaTeX, Markdown, HTML, CSS (with Sass), FastAPI/Streamlit
Soft Skills: Resourceful, Adaptable, Disciplined, Strong Communication, Presentation Skills, Collaboration
Languages: English (native), Hindi (native), Bengali (native), German (A1)

Education

University of Göttingen — Oct 2024 – Oct 2027
PhD Candidate, Data Science — *Göttingen, Germany*

Indian Institute of Science Education and Research (IISER), Pune — Aug 2018 – May 2023
BS-MS Dual Degree (CGPA: 8.1/10, Top 10%) — *Pune, India*

- MS Thesis: *Continual Domain Incremental Learning during Test-time*, GPA 8.7/10.
- Selected Courses: Statistical Inference, Stochastic Processes, Game Theory, Probability, Algorithms and Graphs, Parameter Estimation and Inverse Theory.

Academic Projects

Semester Project: Unsupervised Learning and Bayesian Methods — Jan 2022 – Apr 2022
Indian Institute of Science Education and Research (IISER) — *Pune, India*

- Studied Bayesian inference and clustering algorithms.
- Compared Gibbs Sampling with EM algorithm and KMeans.
- Documented findings and created project site for reproducibility.

Research Intern: Generative Adversarial Networks in Climate Science — Jul 2021 – Jan 2022
Indian Institute of Technology (IITM) — *Pune, India*

- Reviewed GAN architectures applied to climate data.
- Reproduced models such as GAIN, SRGAN, TAnoGAN.
- Explored potential use cases for missing data imputation and anomaly detection.

Summer Intern: Model Predictive Control on COVID-19 Scenario — May 2021 – Sep 2021
Indian Institute of Technology (IIT) - Bombay — *Remote*

- Studied model reinforcement learning, predictive control, mathematical optimization, and economics.
- Framed real-world pandemic control strategies as a control problem. Formulated optimization problems similar to Reinforcement Learning policy search.
- Delivered final report with simulation-based analysis.

Publications

[1] Konstantin Willeke*, Polina Turishcheva*, Alex Gilbert*, **Goirik Chakrabarty** *et al.* 16 more *OmniMouse: Scaling properties of multi-modal, multi-task Brain Models on 150B Neural Tokens.* ICLR, 2026

[2] Pavithra Elumalai, Mohammad Bashiri, **Goirik Chakrabarty**, Suhas Shrinivasan, Fabian H. Sinz. *Beyond Pixels: A Differentiable Pipeline for Probing Neuronal Selectivity in 3D.* NeurIPS-W (NeurReps), 2025 [Oral]

[3] Lubdhak Mondal, Kapil Chandak, **Goirik Chakrabarty**, Indranil SenGupta. *From pixels to profits: a novel approach to identify rare events for a group of US equities.* Annals of Data Science, 2025.

[4] **Goirik Chakrabarty***, Aditya Chandrasekar*, Ramya Hebbalaguppe, Prathosh AP. *LoMOE: Localized Multi-Object Editing via Multi-Diffusion.* ACM Multimedia (ACMMM), 2024.

- **[5]** Aditya Chandrasekar*, **Goirik Chakrabarty***, Jai Bardhan, Ramya Hebbalaguppe, Prathosh AP. *Re-MOVE: A Reference-free Metric for Object Erasure.* CVPR-W (EVGENFM), 2024.
- **[6]** Manogna Sreenivas, **Goirik Chakrabarty**, Soma Biswas. *P-STAR: Pseudo-Source Guided Target Clustering for Fully Test-Time Adaptation.* WACV, 2024.
- **[7]** **Goirik Chakrabarty**, Manogna Sreenivas, Soma Biswas. *SANTA: Source Anchoring Network and Target Alignment for Continual Test-Time Adaptation.* Transactions on Machine Learning Research (TMLR), 2023.
- **[8]** **Goirik Chakrabarty**, Manogna Sreenivas, Soma Biswas. *A Simple Signal for Domain Shift.* ICCV-W (VCL), 2023.
- **[9]** **Goirik Chakrabarty**, Manogna Sreenivas, Soma Biswas. *Domain Shift Signal for Low Resource Continuous Test-Time Adaptation.* ICLR-W (PML4DC), 2023.